Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/23/2007
PI Name:	Varma, Arvind Ph.D.		
Project Title:	Ignition and Combustion of Single Complex Metal Particles Under Reduced Gravity Conditions		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Droplets, sprays, particles, dust clouds, nanoparticles		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	47907-2100	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	12/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Shafirovich, Evgeny Ph.D. (Purdue University)		
Grant/Contract No.:	NNC04AA36A		
Performance Goal No.:			
Performance Goal Text:			

The detailed characterization of metal particle ignition and combustion requires spatial and temporal resolutions that can be achieved only with relatively larger particles (1-5 mm). To avoid the natural convection and liquid flow effects, experiments on combustion of such particles will be conducted in microgravity environment. Since duration of the investigated process is between 5 and 20 s, we plan to conduct experiments by using reduced-gravity research aircraft of NASA. The ignition and combustion will be studied by using a high-speed video camera, thermocouples and product composition analysis. The obtained experimental data will be used to model ignition and combustion of complex metal particles, accounting for characteristic features such as inter-metal reactions, cracking of solid shells, etc. This research is aimed at understanding fundamental problems in combustion of metals related to a number of applications, such as propulsion and power, Mars in-situ resource utilization, and advanced materials synthesis. Specifically, we plan to study combustion of complex particles involving two different metals, one of which forms the core (Al), and the other a thin

Task Description:	shell (Ni, Co, Fe, Cu, etc.). It is important that the particles exhibit high combustibility in both oxidative and inert ambient atmospheres. In the latter case, combustion proceeds due to a high exothermic reaction between the metals, with the formation of solid products (e.g. intermetallics). The main goal of the project is to establish ignition and combustion mechanisms of single complex metal particles in inert (Ar, He) and oxidizer (O2, air, CO2) atmospheres. We plan to determine the ignition (e.g. temperature, delay time) and combustion (e.g. burning rate, flame structure) characteristics of single complex metal particles under different experimental conditions, such as particle composition and size, nature (inert or different oxidizing) of ambient gas. Special attention will be devoted to elucidating the roles of inter-metal reaction and physical processes in surface layers (e.g. cracking of the shell, melt spreading) in the ignition process of complex metal particles. The use of complex metal particles, instead of pure metals, in solid rocket propellants is a promising approach for the enhancement of engine performance due to easier ignition and reduced agglomeration. Further, the decrease in ignition temperature of metal fuel in Mars atmosphere is important for propulsion and power systems using Martian CO2 as an oxidizer.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Combustion of complex metal particles is a novel method for production of a variety of materials, including intermetallics and ceramics. The knowledge obtained from this project will allow development of new technology for synthesis of advanced materials with tailored microstructures and properties.
Task Progress:	The experimental setup for studies on combustion of single complex metal particles under microgravity (parabolic flights) was completed. Structural analysis of the setup was made to confirm that the setup met the NASA requirements. Test Equipment Data Package was prepared and submitted to NASA along with the report on structural analysis. The setup passed safety review by NASA for experiments onboard C-9B aircraft. The ignition of Ni-coated Al particles in Ar and CO2 atmospheres was studied under normal and reduced gravity conditions. The ignition temperature of single Ni-coated Al particles was found to be independent of the atmosphere (CO2 or Ar) and Ni content (provided enough Ni is present). This is the result of a common ignition mechanism, which relies on the sequential formation of NiAl3, Ni2Al3 and NiAl. It was further determined that melting of NiAl3 at 854°C plays a critical role in ignition, leading to self-acceleration of the reaction. Finally, despite the heterogeneous nature of the original particles (Al core and Ni shell), it is remarkable that the product phases are well-mixed. Experiments with Fe-coated Al particles were also conducted and, as a result, some specific features of the Fe coating were revealed. Experiments in microgravity during parabolic flight were performed using both Ni-coated and Fe-coated Al particles. In total, 84 experiments were successfully completed, and analysis of the results is in progress. The work supported by this grant has been described in four journal articles, including two invited, and presented at five scientific conferences.
Bibliography Type:	Description: (Last Updated: 01/05/2007)
Abstracts for Journals and Proceedings	Andrzejak TA, Shafirovich E, Varma A. "Ignition mechanisms of metal-coated aluminum particles." 2006 AIChE Annual Meeting, San Francisco, CA, November 12-17, 2006. 2006 AIChE Annual Meeting, November 2006. http://www.aiche.org , Nov-2006
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Papers from Meeting Proceedings	Andrzejak TA, Shafirovich E, Taylor DG, Varma A. "Studies on ignition and combustion mechanisms of single Ni-coated Al particles." 44th AIAA Aerospace Sciences Meeting and Exhibit, Reno, NV, January 9-12, 2006. AIAA Paper 2006-1130. January 2006. , Jan-2006
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Presentation	Andrzejak, T.A.; Shafirovich, E.; Taylor, D.G.; Varma, A.; "Studies on Combustion of Single Ni-Coated Al Particles in Normal and Reduced Gravity" 2005 AIChE Annual Meeting, Cincinnati, OH, October 30-November 4, 2005 Nov-2005